Willmer Rafell Quiñones Robles

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https://wrafell.github.io/

Research Interest

- I am broadly interested in developing and applying deep learning-based systems to advance the fields of healthcare, visual recognition, and cognitive science. My work focuses on building robust and efficient representation learning models that are both sample-efficient and aligned with the way humans learn and perceive the world. I am also interested in the application of federated learning to enable privacy-preserving model training in real-world clinical settings.
  Keywords: AI for Healthcare, Computational Pathology, Deep Representation Learning, Self-Supervised Learning, Whole-Slide Image Analysis, Federated Learning, Cognitive-Inspired Learning, Visual Recognition.

Education

Feb 2020 – Feb 2025
- **Ph.D., Graduate School of Data Science**
  Korea Advanced Institute of Science and Technology (KAIST) · Advisor: Mun Young Yi
  Area of Study: Machine Learning in Healthcare, Deep Learning for Computer Vision, Self-Supervised Learning
  Thesis: *Spatial Context-Driven Approach for Efficient Cancer Classification in Whole-Slide Images*

Feb 2018 – Feb 2020
- **M.S., Graduate School of Knowledge Service Engineering**
  Korea Advanced Institute of Science and Technology (KAIST) · Advisor: Mun Young Yi
  Area of Study: Machine Learning in the Healthcare field, Deep Learning for Computer Vision
  Thesis: *Impact of Cancer Histopathological Image Preprocessing on Convolutional Neural Network Performance: A Sensitivity Analysis*

Feb 2011 – Aug 2014
- **B.S., Electronic and Communication Engineering**
  Santo Domingo Institute of Technology (INTEC)
  Graduated with Honors · GPA 3.55 / 4

Professional Experience

Oct 2025 – Present
- **Postdoctoral Research Associate,** Lab of Advanced Imaging Technology (LAIT), UNIST, South Korea.
  Supervised by Prof. Jaejun Yoo.

Mar 2018 – Feb 2025
- **Research Assistant,** Knowledge Innovation Research Center (KIRC) @ KAIST, Daejeon, South Korea.
  Diagnostic Pathology using Deep Learning · Develop Deep Learning models to automatically diagnose abnormalities on histopathology images.

Feb 2021 – Oct 2024
- **Course Instructor,** Santo Domingo Institute of Technology (INTEC), Dominican Republic (Remote).
  Served as the instructor for undergraduate courses, including Introduction to Artificial Intelligence, Artificial Intelligence, and Deep Learning.

Professional Experience (continued)

Feb 2015 – Jan 2018 — **SCADA System Engineer,** Dominican Electrical Transmission Company (ETED), Dominican Republic.
Managed the Supervisory Control and Data Acquisition system that covers all the electrical transmission power lines in the Dominican Republic.

Publications

Journal Articles

1. S. Noree, **W. R. Quinones Robles**, Y. S. Ko, and M. Y. Yi, "Leveraging commonality across multiple tissue slices for enhanced whole slide image classification using graph convolutional networks," *BMC Medical Imaging*, vol. 25, no. 1, p. 11, 2025.
2. M. Kim, **W. R. Quinones Robles**, Y. S. Ko, B. Wong, S. Lee, and M. Y. Yi, "A predicted-loss based active learning approach for robust cancer pathology image analysis in the workplace," *BMC Medical Imaging*, vol. 24, no. 1, p. 5, 2024.
3. M. Ashraf, **W. R. Quinones Robles**, M. Kim, Y. S. Ko, and M. Y. Yi, "A loss-based patch label denoising method for improving whole-slide image analysis using a convolutional neural network," *Scientific Reports*, vol. 12, no. 1, p. 1392, 2022.
4. Y. S. Ko et al., "Improving quality control in the routine practice for histopathological interpretation of gastrointestinal endoscopic biopsies using artificial intelligence," *PLOS ONE*, vol. 17, no. 12, e0278542, 2022.

Conference Proceedings

1. J. W. Kim, B. Wong, H. Fu, **W. R. Quinones Robles**, Y. S. Ko, and M. Y. Yi, "MicroMIL: Graph-based multiple instance learning for context-aware diagnosis with microscopic images," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2025.
2. **W. R. Quinones Robles**, M. Ashraf, and M. Y. Yi, "Impact of patch extraction variables on histopathological imagery classification using convolution neural networks," in *2021 International Conference on Computational Science and Computational Intelligence (CSCI)*, IEEE, 2021, pp. 1176–1181.

Preprints / Under Review

1. **W. R. Quinones Robles**, S. Noree, Y. S. Ko, B. Wong, J. W. Kim, and M. Y. Yi, "Leveraging spatial context for positive pair sampling in histopathology image representation learning," arXiv preprint, 2025.
2. **W. R. Quinones Robles**, S. Noree, Y. S. Ko, and M. Y. Yi, "Artificial class activation maps using fractals: A new data augmentation strategy for deep learning-based whole-slide image analysis," Under Review, 2024.

Teaching Experience

Aug 2024 – Dec 2024 — **Teaching Assistant,** Scientific Writing for AI, KAIST (Remote).
Attendees: Graduate Students of the Graduate School of Artificial Intelligence, KAIST.

Feb 2024 – Jun 2024 — **Teaching Assistant,** Scientific Writing, KAIST (Remote).
Attendees: Graduate Students at KAIST.

Teaching Experience (continued)

May 2024 – Oct 2024 — **Lecturer,** Artificial Intelligence, INTEC, Dominican Republic (Remote).
Teaching Artificial Intelligence focused on Software Engineering.
Attendees: Undergraduate Students at Santo Domingo Institute of Technology (INTEC).

Feb 2021 – Apr 2024 — **Lecturer,** Introduction to Artificial Intelligence, INTEC, Dominican Republic (Remote).
Teaching general topics of Artificial Intelligence.
Attendees: Undergraduate Students at Santo Domingo Institute of Technology (INTEC).

Aug 2022 – Oct 2022 — **Lecturer,** Introduction to Deep Learning, INTEC, Dominican Republic (Remote).
Teaching introductory topics of Deep Learning.
Attendees: Undergraduate Students at Santo Domingo Institute of Technology (INTEC).

Skills

Research — Deep learning, computational pathology, whole-slide image analysis, medical image computing, visual recognition, data augmentation, multiple instance learning, self-supervised representation learning, federated learning, cognitive-inspired learning.

Programming — Python (PyTorch, JAX), R, C/C++, MATLAB, LaTeX.

Languages — Spanish (native), English (fluent), Korean (intermediate).

Scholarships

Feb 2018 – Feb 2024 — **Full Academic Scholarship,** KAIST.

Jul 2021 – Oct 2021 — **Global AI & Big Data Scholarship Program,** Daewoong Foundation.

Jan 2021 – Apr 2021 — **Global AI & Big Data Scholarship Program,** Daewoong Foundation.

Feb 2018 – Feb 2020 — **KAIST-KOICA-MESCYT Project.**

References

Available on Request